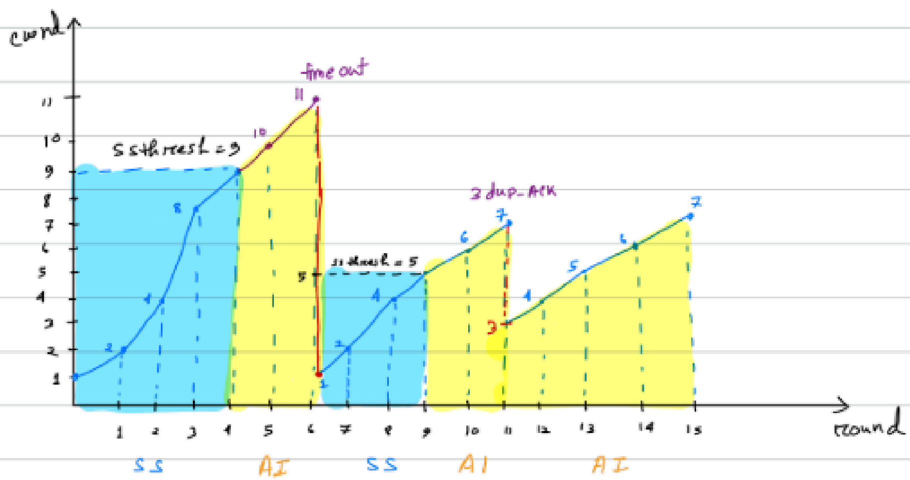
Solution of Congestion Control Math

1. A file transfer starts at MSS=1; ssthresh=9; round=0. The system faces a time-out at around 6 and then a 3-dup-ack on round 11.
    a. Identify the MSS size on round 15.
    b. Identify the ssthresh on round 15.
    c. Total of how many bytes were sent till round 15 if each MSS held 3 bytes of data?



a. MSS size on round 15 is 7.

b. ss threshold will be :

i. For timeout at 6
so, $11 \div 2 = \lfloor 5.5 \rfloor = 5$

ii. For 3 ACK at round 11
$ssthresh = 7 \div 2 = \lfloor 3.5 \rfloor = 3$ (Ans)

c. Total mss's $= 1 + 2 + 4 + 8 + 9 + 10 + 11 + 1 + 2 + 4 + 5 + 6 + 7 + 3 + 4 + 5 + 6 + 7$

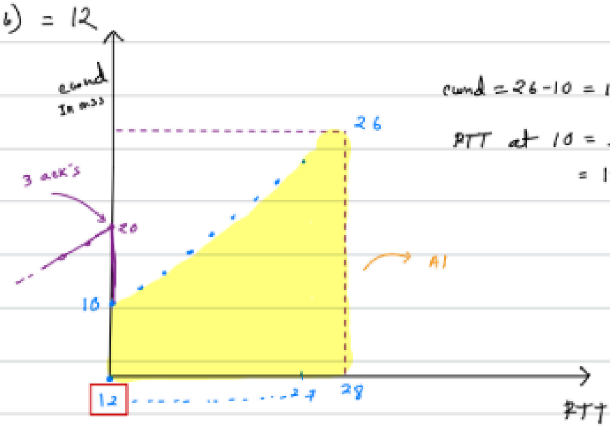
$= 95$

Now, total bytes $= 95 \times 3 = 285$ bytes

2. A file transfer is currently at MSS = 26; ssthresh = 10, rtt = 28
   a. Identify if there was any congestion along the transmission.
   b. If the last congestion occurred due to 3 ACKs, what was the value of RTT at that time?
   c. If the last congestion occurred due to time-out, what was the value of RTT at that time?

a. Here mss is greater equal than ssthresh.
So, there will be congestion along the transmission.

b. ss threshold = 10, RTT = 28, mss = 26
If last congestion occurred due to 3 acks, means after that mss increased by 1 each RTT.
So, RTT is (28-16) = 12



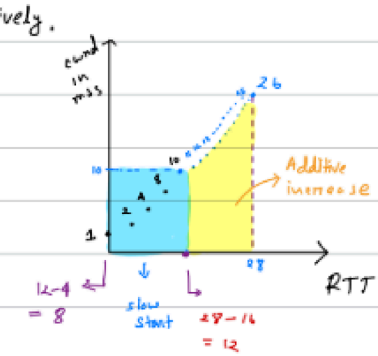
cwnd = 26-10 = 16;
RTT at 10 = 28-16
= 12

c. If the last congestion occurred due to timeout
So, After timeout the will be slow start means each round cwnd will increase multiplicatively.

when cwnd is 10, RTT = 28 − (26 − 10)
= 12

Before that was slow sdart.

so, at cwnd = 1, RTT = (12 − 4)
= 8 (Ans)



3. In TCP Congestion Control, the sender started (at RTT=0) with the slow start and the initial ssthresh was set to 16. Then the following events occurred.

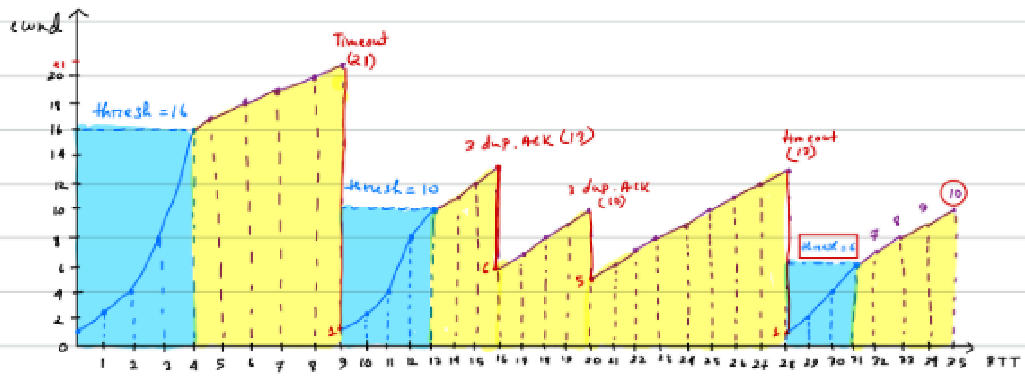
- Event 1: Time-out occurred at RTT = 9
- Event 2: 3 Duplicate acks received at RTT = 16
- Event 3: 3 Duplicate acks received at RTT = 20
- Event 4: Time-out occurred at RTT = 28

a. Draw the cwnd vs rtt graph for the above scenario.

b. Calculate the value of ssthresh at RTT=35.

c. Calculate the value of cwnd at RTT=35.

a.



b. At RTT 35, value of ssthresh is 6.

c. At RTT 35, value of cwnd is 10.